Folio state in guest_memfd

Host: Determines who can map/fault-in a folio.

Mappability {ALL, GUEST, NONE}

- ALL: Mappable by host and guest (Shared)
- GUEST: Mappable only by guest (Private)
- NONE: Not mappable by either host or guest, transitioning from Private -> Shared.
  Note: There could be existing mappings, but any new mappings will fail.

Guest S2: Guest hypervisor page state : {NOPAGE, OWNED, SHARED}

These states are not explicitly defined as such, but can be deduced by the host.

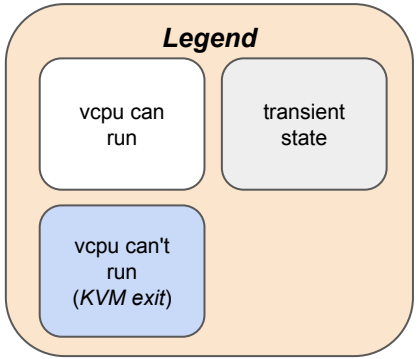
- NOPAGE: The page has not been faulted into the guest
- OWNED: The page is private to the guest
- SHARED: The page is shared with the host

Refcount: The host view of the refcount (folio_ref_count())

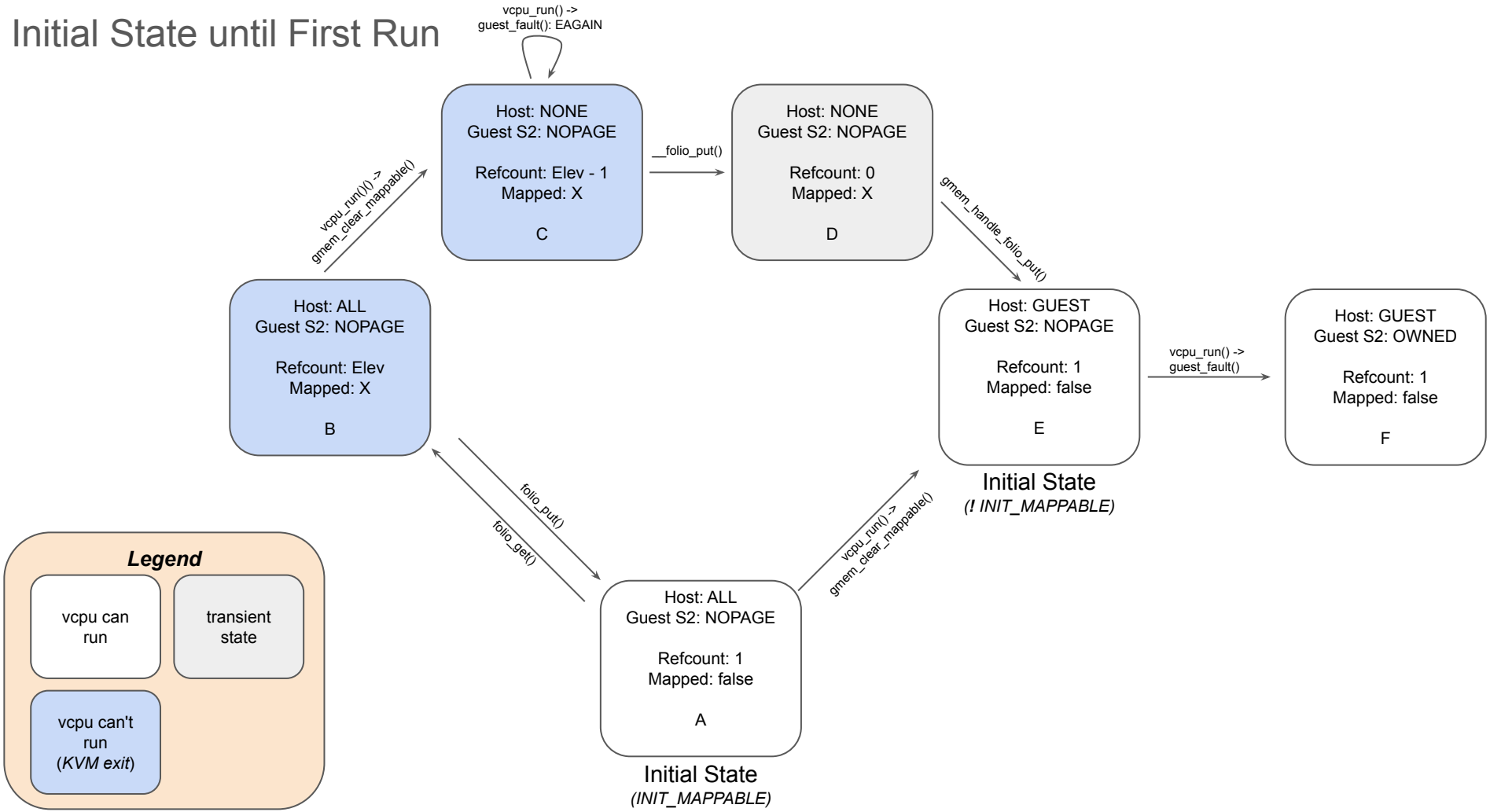
`folio_ref_count()`: {0, 1, Elev (> 1), Elev - 1}
- 0: The refcount is 0. A transient state that happens to trigger __folio_put() callback.
- 1: The refcount is the base refcount for the folio. For simplicity, folio is one page.
- Elev: refcount is elevated, with a higher value than what can be accounted for.
- Elev - 1: refcount is elevated, but we have subtracted base count to get a callback.

Mapped: The host view of the mapcount (folio_mapped())

- false: No userspace mappings (_mapcount == 0)
- X: Do not care



Initial State until First Run



Share and Unshare

